AI4I 2020 Predictive Maintenance Classification

Executive summary

Four classifiers were evaluated on the same stratified 2,000-row held-out test set. RBF-SVM has the highest held-out ROC-AUC (0.9767); Decision Tree has the highest failure-class F1 (0.7846). The CNN has the highest failure recall (0.8824) at threshold 0.5. The results show that model choice depends on whether ranking quality, balanced failure detection, or missed-failure reduction is prioritized.

Dataset and task

The UCI AI4I 2020 Predictive Maintenance dataset contains 10,000 rows. The target is Machine failure, with label 1 representing failure. The positive class is rare, around 3.4 percent. Inputs are product Type and five sensor measurements: air temperature, process temperature, rotational speed, torque, and tool wear.

Leakage controls and preprocessing

UDI and Product ID are excluded as identifiers. TWF, HDF, PWF, OSF, and RNF are excluded because they are failure-mode labels that leak target information. KNN and RBF-SVM use one-hot Type and standardized numeric values. Decision Tree uses ordinal Type (L < M < H) and raw numeric values. CNN uses one-hot Type and standardized numeric values; its preprocessor is fit only on its 7,200-row fit subset.

CNN interpretation

The CNN is a Conv1D model over the eight preprocessed tabular features in a fixed order, shaped as (8, 1). Adjacent columns do not have a guaranteed local relationship. Instructor acceptance of this tabular CNN interpretation has not been confirmed.

Experimental design and model selection

Shared outer split

All four models use the same stratified 80/20 split with random seed 42: 8,000 training rows and 2,000 held-out test rows. The ordered test indices saved in the three classical metadata files match the CNN metadata. Test rows were not used for hyperparameter selection or CNN checkpoint selection.

Classical cross-validation

KNN, Decision Tree, and RBF-SVM use stratified 5-fold cross-validation on training rows. GridSearchCV selects the candidate with the highest failure-class F1 and refits it on all training rows. Preprocessing is inside each pipeline. Best settings and CV F1 values below are read from the saved metadata and CV result tables.

CNN validation and checkpointing

The CNN separates 800 validation rows from the 8,000-row training portion, leaving 7,200 fit rows. Class weights are calculated from fit labels only. Early stopping, learning-rate reduction, and checkpointing monitor validation PR-AUC. The saved best checkpoint is epoch 12 of 22; validation PR-AUC is 0.7742 and validation ROC-AUC is 0.9830.

Metric interpretation

Precision, recall, and F1 treat failure (1) as positive. Thresholded predictions use the saved threshold of 0.5. ROC-AUC and average precision use continuous positive-class scores: sigmoid probabilities for CNN and decision_function scores for SVM. Metrics are unweighted. The CNN's class-weighted training loss is not directly comparable to unweighted validation/test loss.

Saved model results

Selected classical hyperparameters and CV failure-class F1

Model	Selected parameters	Mean CV F1 ± std
KNN	n_neighbors=1, weights=uniform	0.4928 ± 0.0731
Decision Tree	class_weight=None, max_depth=10, min_samples_leaf=5	0.7245 ± 0.0528
RBF-SVM	C=100, class_weight=None, gamma=scale	0.6912 ± 0.0182

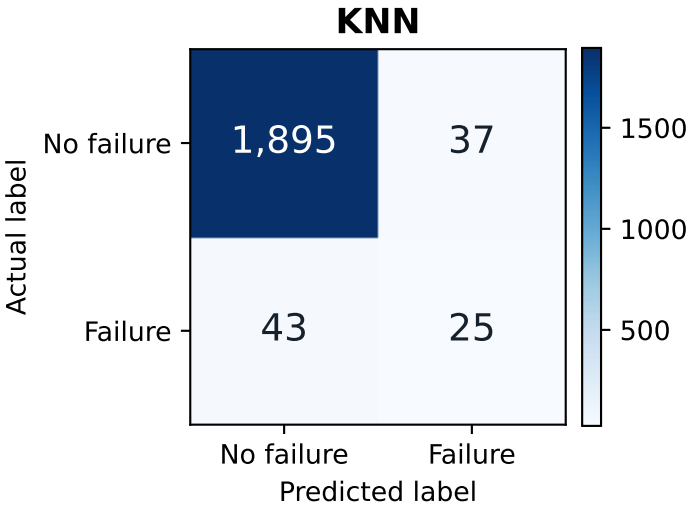
Held-out test metrics (2,000 shared rows)

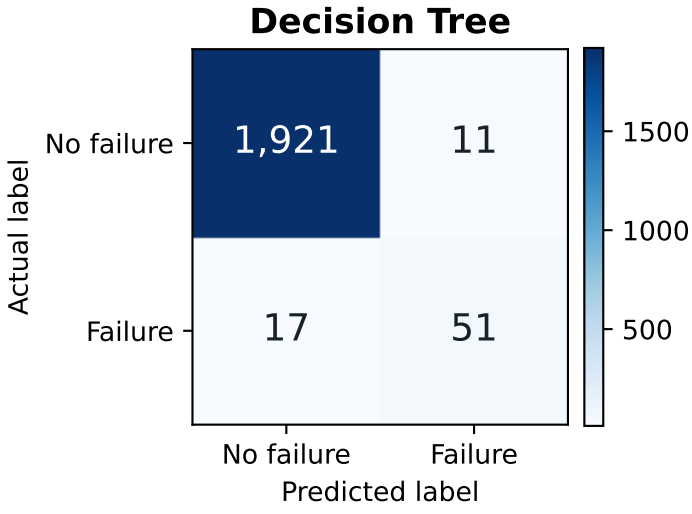
Model	Accuracy	Precision	Recall	F1	ROC-AUC	Avg. precision
KNN	0.9600	0.4032	0.3676	0.3846	0.6742	0.1697
Decision Tree	0.9860	0.8226	0.7500	0.7846	0.8910	0.7453
RBF-SVM	0.9800	0.7593	0.6029	0.6721	0.9767	0.7862
1D CNN	0.9340	0.3261	0.8824	0.4762	0.9714	0.7242

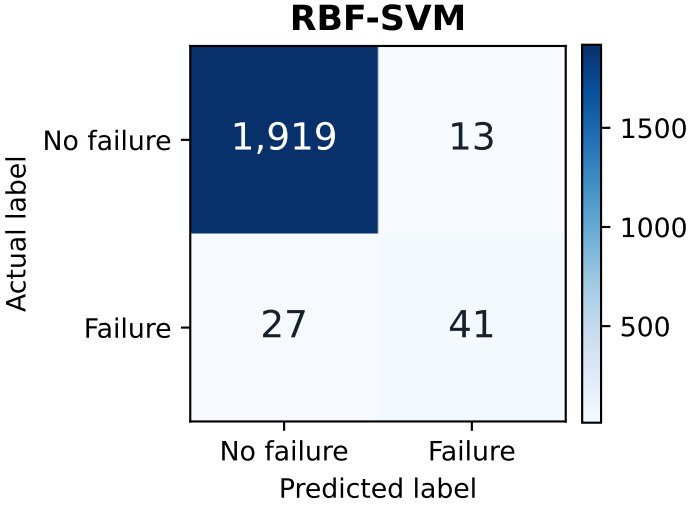
Confusion counts (failure class 1 is positive)

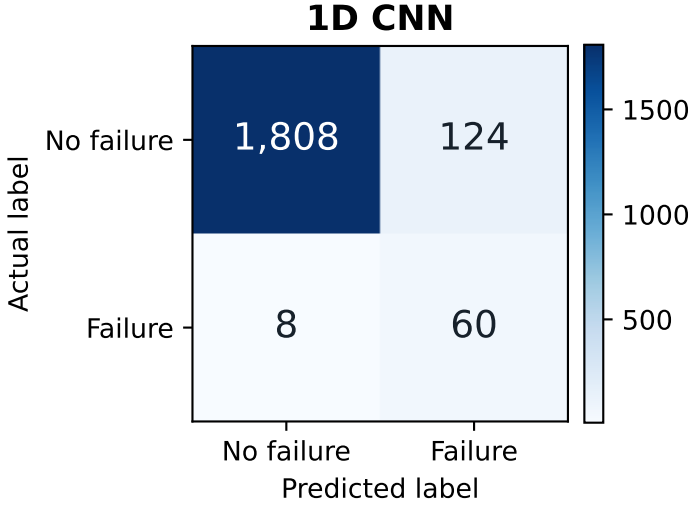
Model	TN	FP	FN	TP
KNN	1,895	37	43	25
Decision Tree	1,921	11	17	51
RBF-SVM	1,919	13	27	41
1D CNN	1,808	124	8	60

Held-out confusion matrices



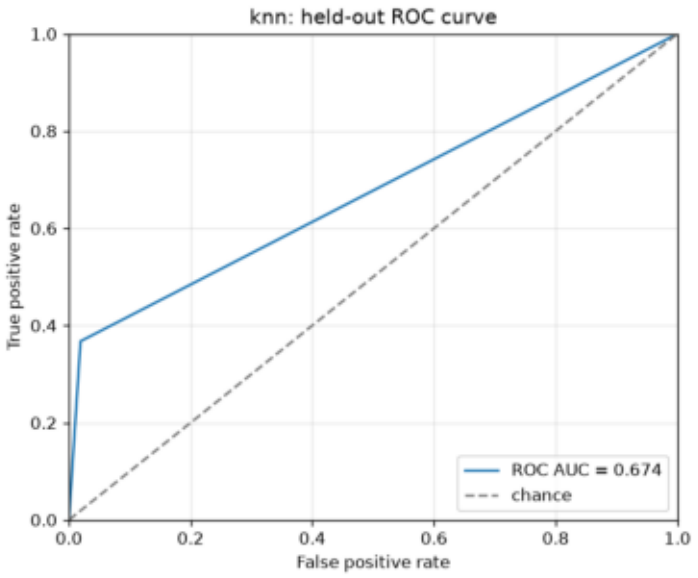




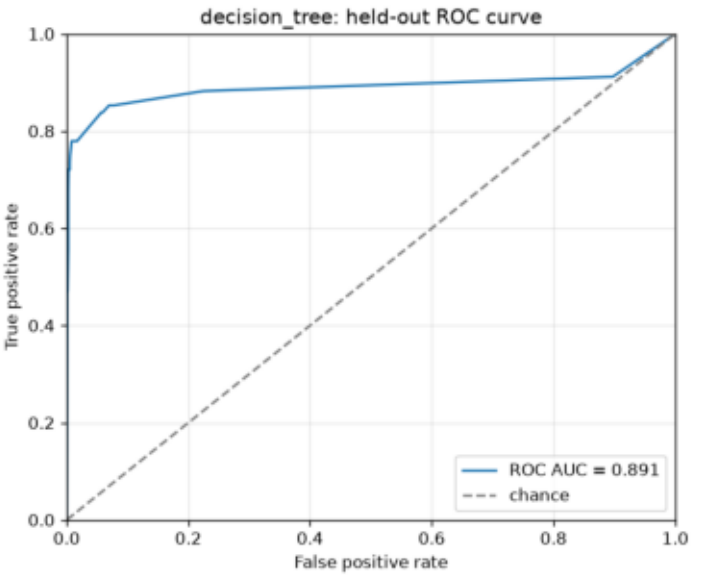


Held-out ROC curves

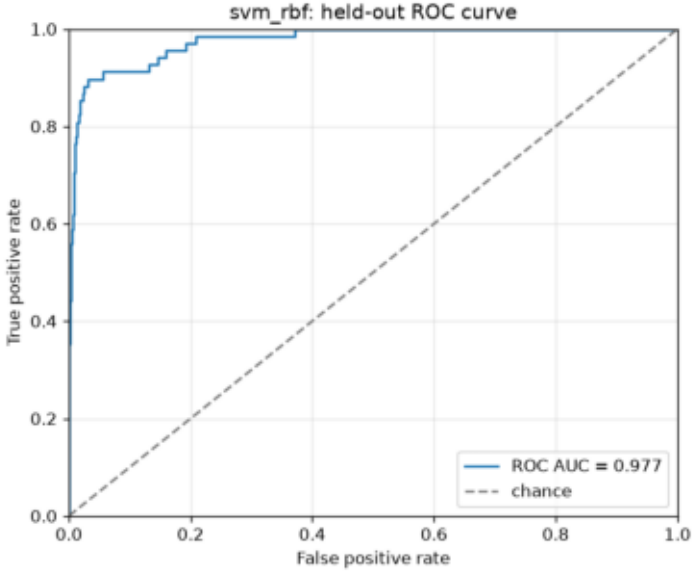
KNN — ROC-AUC 0.6742



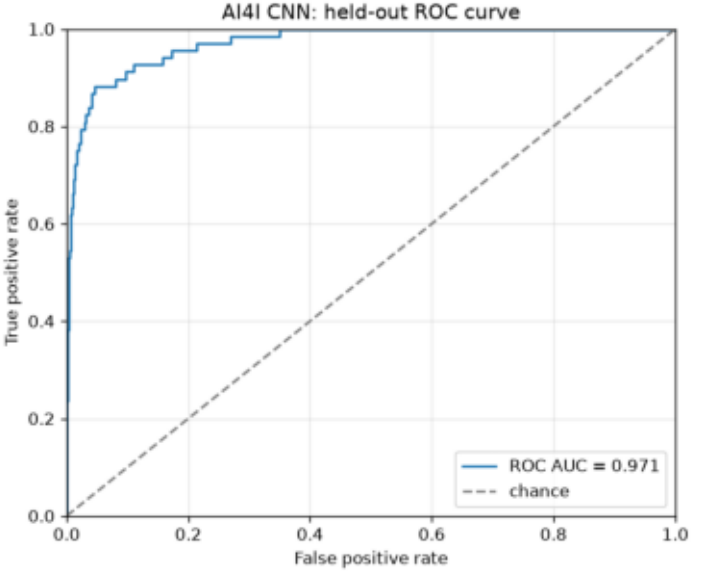
Decision Tree — ROC-AUC 0.8910



RBF-SVM — ROC-AUC 0.9767



1D CNN — ROC-AUC 0.9714



Generalization and overfitting

KNN

At its selected candidate, mean training-fold F1 is 1.0000, versus mean CV F1 0.4928, a gap of 0.5072. Test F1 is 0.3846. The large gap and lower held-out result are clear evidence of a generalization concern for the selected 1-neighbor model.

Decision Tree and RBF-SVM

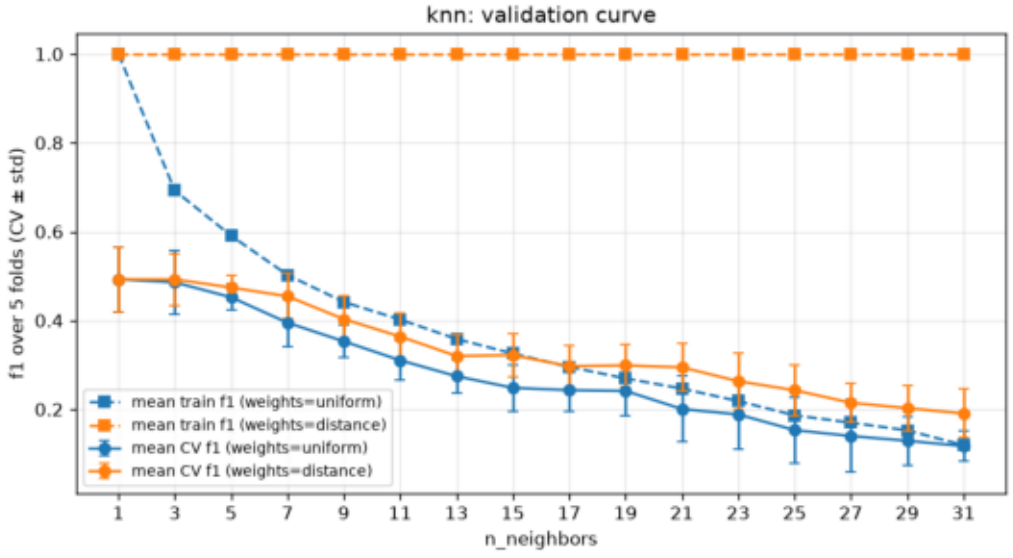
Decision Tree mean training-fold F1 is 0.8269 and mean CV F1 is 0.7245; held-out F1 (0.7846) is above its CV mean. RBF-SVM mean training-fold F1 is 0.8721 versus CV F1 0.6912; held-out F1 (0.6721) is close to CV. The training/CV gaps warrant attention but the held-out results do not show a large additional decline.

CNN

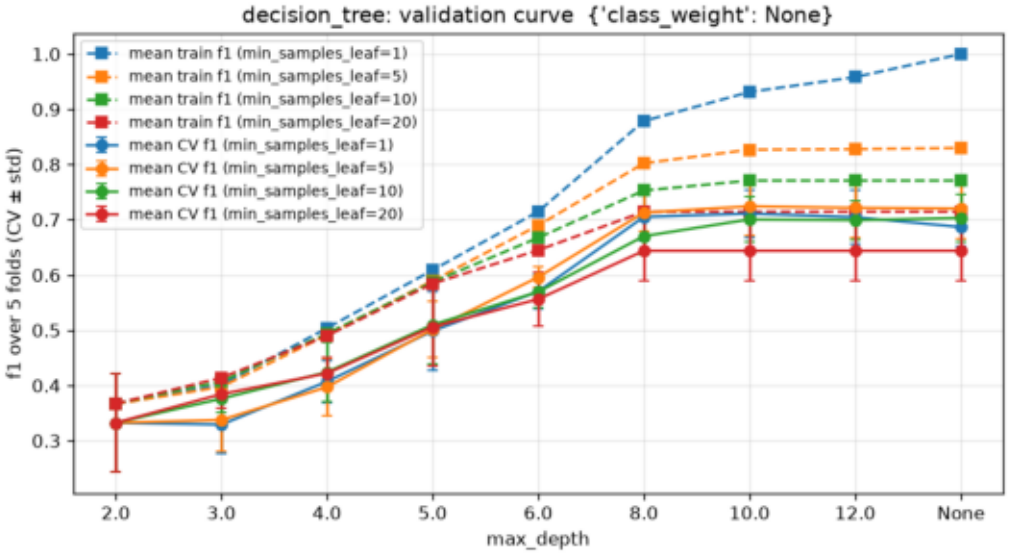
CNN validation PR-AUC peaked at 0.7742 at epoch 12. By epoch 22, training PR-AUC reached 0.7681 while validation PR-AUC was 0.6806; validation ROC-AUC changed from 0.9830 to 0.9759. This supports late-stage overfitting; the saved checkpoint is the best validation epoch. Training loss is class weighted and validation loss unweighted, so the loss curves should not be compared directly.

Training and validation diagnostics

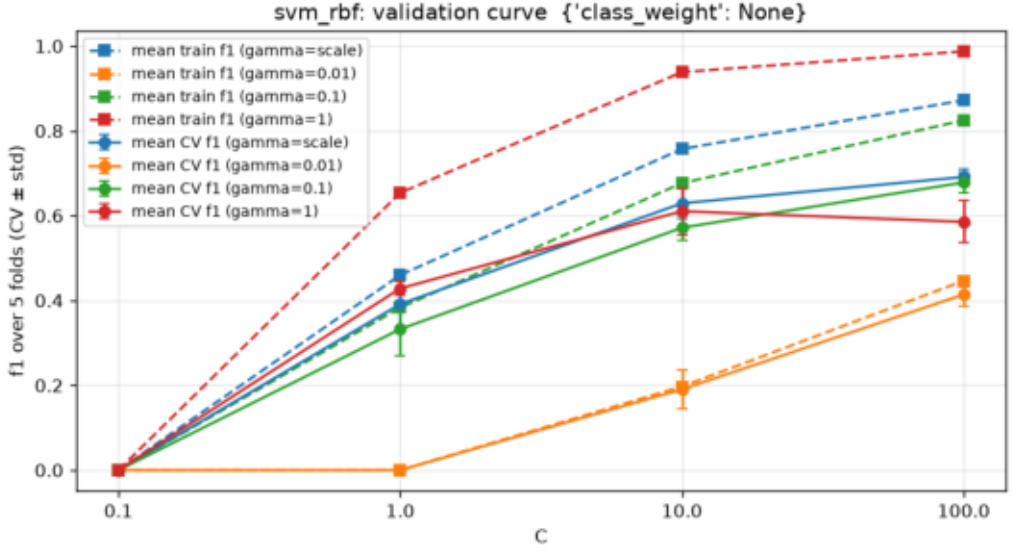
KNN: training-fold and CV F1



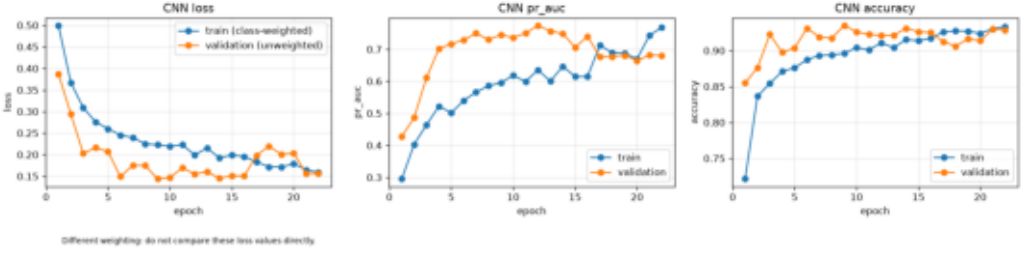
Decision Tree: training-fold and CV F1



RBF-SVM: training-fold and CV F1



CNN: training and validation history



Limitations, conclusion, and reproducibility

Limitations

The failure class is rare, and this report summarizes one fixed train/test split. No repeated-split uncertainty analysis or statistical significance testing is included. A 1D CNN assumes relationships between adjacent input positions even though the features are tabular and their order is arbitrary; instructor acceptance of this interpretation is unresolved. The CNN's threshold is fixed at 0.5 and was not optimized on held-out test data.

Conclusion

RBF-SVM provides the strongest held-out ranking performance by ROC-AUC (0.9767), while Decision Tree achieves the highest failure-class F1 (0.7846). CNN recall is highest (0.8824) but comes with lower precision (0.3261) and many more false positives. The preferred model therefore depends on the cost of missed failures versus false alarms; the available results do not establish a deployment choice.

Reproducing the saved experiment

From the project root, run `python -m classification.train` for classical tuning/training, `python -m cnn.train` for CNN training, and `python -m cnn.evaluate` to evaluate the saved CNN checkpoint. Training commands rerun experiments and overwrite outputs. Run `python -m test_member1` and `python -m unittest -v test_classification_evaluation test_cnn_evaluation test_validation_curve_plotting` for checks. This report was generated from saved results; models were not retrained.

Data citation and open submission question

Dataset: AI4I 2020 Predictive Maintenance Dataset, UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset. No assignment rubric was present in the repository. Confirm the required report format and whether the instructor accepts the tabular 1D CNN interpretation before final submission.